

**What is****THERMODYNAMICS**

Thermodynamics is the science that deals with the relationship between heat and other forms of energy. Earlier, in 1824, the French engineer, **Nicolas Sadi Carnot**, published the "Reflections on the Motive Power of Fire", a discourse on heat, power and engine efficiency. Later, he became known as the "**Father of Thermodynamics**".



The word "thermodynamics" comes from the Greek words "therme" and "dynamics", which means "heat" and "power", respectively. This term was coined by the English physicist, **James Prescott Joule**, in 1849.

Excel Review Center



In 1824, **Nicolas Sadi Carnot**, published the "Reflections on the Motive Power of Fire", a discourse on heat, power and engine efficiency. Later, he became known as the "**Father of Thermodynamics**".

Branches of Thermodynamics

1. **Classical Thermodynamics** - It is the branch of thermodynamics that is concerned with thermodynamic states and properties (energy, work and heat) and with the laws of thermodynamics without the atomic interpretation. This is the macroscopic approach of thermodynamic analysis.
2. **Statistical Thermodynamics** - It is the branch of thermodynamics that includes the atomic and molecular theories. It is the microscopic approach of thermodynamics.
3. **Chemical Thermodynamics** - It is the branch of thermodynamics that is concerned with the study of the interrelation of heat with chemical reactions or with a physical change of state within the confines of the laws of thermodynamics.

Excel Review Center



The Future of Education!

Like us on Facebook
alterguru.com**Phases of Substances**

Normally, there are three phases of a substance, i.e. **solid**, **liquid** and **gas**, along with two added phases, i.e. **plasma** and **Bose-Einstein condensate**. This was discussed in Chemistry. Since different substances have different properties and behave differently, the following are the more specific phases of substances:

Excel Review Center

1. **Solid** - In this phase, the substance does not take the shape or volume of the container.
2. **Subcooled liquid** - In this phase the liquid is non-saturated. Non-saturated means that it is not at its boiling point.
3. **Saturated liquid** - In this phase the liquid that can absorb as much heat as it can without vaporizing.
4. **Liquid-vapor mixture** - In this phase, the liquid and vapor co-exist with the same temperature and pressure.
5. **Saturated vapor** - In this phase, the vapor has absorbed more heat than necessary to vaporize it.
6. **Ideal gas** - In this phase, the gas (a highly superheated vapor) behaves in accordance with the ideal gas law.
7. **Real gas** - In this phase, the gas does not behave in accordance with the ideal gas law.
8. **Gas mixtures** - In this phase, two or more gases mixed together freely.
9. **Vapor/Gas mixtures** - In this phase, two or more gases mixed freely with water vapor.

Excel Review Center

Temperature

Temperature is defined as the degree of hotness or coldness of a certain substance with respect to some standard value. It is a property that is dependent on the substance's energy content. To be more specific, it is a measure of the average kinetic energy of the different particles in a sample of matter. A **thermometer** is a device that measures temperature or a temperature gradient. A thermometer has two important elements: a temperature sensor in which some change occurs with a change in temperature; and some means of converting this change into a numerical value.



Analog thermometer



Digital thermometer

Excel Review Center

Temperature Scales

The 4 common temperature scales used are:

1. **Celsius scale** - Also known as the Centigrade scale, this is perhaps the most widely accepted temperature scale used globally. In this scale, there are 100 degrees of separation between the boiling point and the freezing point of water, hence the word "centigrade", which literally means "100 steps". This temperature scale is named after the Swedish astronomer, **Anders Celsius**.
2. **Fahrenheit scale** - Named after the German physicist, **Gabriel Daniel Fahrenheit**, it is another common scale used in some countries most notably in the United States. In this scale, there are 180 degrees of separation between the boiling point and the freezing point of water. Fahrenheit was the inventor of the first mercury thermometer that we now use today.
3. **Kelvin scale** - This is the SI unit of temperature and of course, the most common unit of temperature adopted in the scientific community. It is similar to the Celsius scale wherein there are 100 degrees of separation between the boiling point and the freezing point of water. However, the difference lies in the fact that the 0 Kelvin is 273 units colder than its equivalent value in the Celsius scale. 0 Kelvin is called absolute zero temperature. It is the lowest temperature that any matter can reach or it is the temperature below which any other value has no physical significance. This temperature scale is named after the British physicist named **Sir William Thomson** or **Lord Kelvin**. In writing values of temperature in the Kelvin scale, the symbol K is used without the degree symbol.
4. **Rankine scale** - Named after **William John Rankine**, this is the absolute scale equivalent of the Fahrenheit scale. This is also similar to the Fahrenheit scale wherein there are 180 degrees of separation between the boiling point and the freezing point of water.

Excel Review Center